MANIVANNAN S

Data Analyst / Data Engineer
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PROFESSIONAL SUMMARY

Data engineer and analyst with 3 years of experience in the renewable energy industry, working with wind turbine and solar telemetry. Builds Python pipelines that take data from SCADA, PLCs and IoT devices and turn it into reports, dashboards, alerts and long-term archives for engineering and service teams. I work with AWS (S3, EC2), SQL and ThingsBoard, and I've led a production platform migration with minimal downtime. I'm now building AI-driven tools that let operators query turbine data in plain language.

CORE SKILLS

- **Programming:** Python, Pandas, Numpy, Advanced SQL (joins, window functions), PLC,
- **Data Analytics:** Power BI, Tableau, Exploratory Data Analysis (EDA), KPI reporting, Data Visualization, SCADA, Time Series & IoT Data.
- **Data Engineering:** ETL Pipelines, Data migration, Data Archival, Data Quality, Historical Data management.
- **Cloud & Infrastructure:** AWS EC2, Amazon S3, Linux, Docker, Thingsboard PE.
- **AI & Machine Learning**: RAG, LLM, Scikit-learn, Tensorflow, Pytorch.
- **Application Development:** Flutter, Dart, REST APIs, HTML, CSS, JavaScript.

EXPERIENCE

PIONEER WINCON ENERGY SYSTEMS PVT LTD Chennai, TN
December 2023 – Present

Data Analytics

- Created Power BI and Tableau dashboards that let engineering and service teams track turbine generation, performance indicators and other operating parameters.
- Automated the weekly reports on power generation, turbine breakdowns and irradiance, saving over 20 hours of manual work every week.
- Wrote Python tools that extract PLC faults and pinpoint turbine failures at 20 ms resolution, cutting diagnostic time by 50%. Used them to analyze faults in hydraulic units, converters, main bearings, generators and gearboxes.
- Built a Telegram fault-alert system on the ThingsBoard APIs so teams hear about turbine faults right away and can prioritize troubleshooting. Turbine downtime fell by a reported 35% after rollout.
- Made Python tools for power-curve analysis and generation reports, so the O&M team can pull operational data without waiting on the analysis team.

Data Engineering

- Built Python ETL workflows that collect, clean and organize wind turbine and solar telemetry for operational reporting and performance analysis.
- Developed a Python-based pipeline (buffering) on Raspberry Pi and IoT devices that stores turbine and solar telemetry in local database and automatically sends any unsent records to the production server once the internet connection is restored, achieving 100% data availability.
- Built an archival pipeline that moves historical telemetry to Amazon S3 and applies FIFO retention. Over two years of data stays accessible while database storage stays under control.
- Migrated the production ThingsBoard platform from 2.4.1 to 4.3.1.2, including databases, dashboards, users, rule chains and configurations, with about two hours of downtime.
- Set up and tested RF networks linking remote turbines, keeping SCADA data and remote monitoring online.
- Built an authenticated, offline web app (HTML, CSS, JavaScript, Python) that lets operators curtail solar power over the RF network, much faster than the earlier manual process.

CURRENT PROJECTS

AI Turbine Operations Assistant and SQL Chatbot | Python, SQL, RAG, LLMs, APIs

- Built a chatbot that turns plain-English questions into database queries and returns the results in a readable format, so operators and analysts get answers without writing SQL.
- Now extending it into an assistant for turbine operators and customers. It will cover turbine status, active alarms, generation reports and performance data, and use RAG to give troubleshooting guidance.

ThingsBoard Mobile Application | Flutter, Dart, REST APIs, ThingsBoard PE

- Building a Flutter mobile app that gives site users and customers direct access to their ThingsBoard dashboards through REST APIs.
- Covers live turbine status, active alarms and historical telemetry, so users can check their turbines without logging in to the web platform on a desktop.

PERSONAL PROJECTS

Fake Number Plate Detection | Python, YOLO, CNN, Deep Learning | March 2024

- Designed a web-based system that uses YOLO and CNN models to extract vehicle details from video, including number plate, make and model, color and travel trajectory
- Designed to flag fake plates with a planned alert to the nearest police unit.

IPL Player Auction Prediction | Python, Web Scraping, XGBoost, Random Forest, Decision Tree | July 2023

- Scraped and cleaned 14 years of IPL data with Python, then built three machine learning models for the auction
- Used XGBoost regression to predict a player's auction price, a Random Forest classifier to predict the buying team, and a Decision Tree classifier to predict whether the player is sold or unsold.

EDUCATION

M.Sc. in Data Science with Business Analysis | July 2022 to May 2024 | MSc final-year project

Rathinam College of Arts and Science, Coimbatore, Tamil Nadu

CGPA: 8.19/10

Coursework: Data Analysis, Data Visualization, Machine Learning, Deep Learning, Big Data Analytics

Bachelor of Computer Applications (BCA) | July 2018 to September 2021

St. Thomas College of Arts and Science, Chennai, Tamil Nadu

CGPA: 7.7/10

Coursework: Database Management Systems, Data Structures and Algorithms, Operating Systems, Software Engineering

CERTIFICATIONS

- Google Advanced Data Analytics
- Microsoft AI & ML Engineer
- IBM Data Engineer
- Cloud Data Management (Oracle)
- Power BI (Simplilearn)
- Advance SQL (HackerRank)

PUBLICATION & ACTIVITIES

- Published a conference paper, Recognizing and Intercepting Users of Fake Number Plates, at an international conference.
- Took part in Smart India Hackathon 2022, a Government of India national innovation competition. Our team proposed an ML, DL and NLP-based platform for farmers, offering soil-based cultivation guidance, weather reports, a natural-language knowledge library and a marketplace.